

Climate Analytics Project

Topic: Impact of climate change on tourism in the Coral Reef in Australia

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Introduction:

Background and Motivation:

Climate change represents one of the most pressing global challenges of the modern era, with far-reaching consequences for natural ecosystems, economic systems, and human well-being. Marine ecosystems are particularly vulnerable to climate-induced stressors such as rising sea surface temperatures, increased frequency of extreme weather events, and declining water quality. Among these ecosystems, coral reefs are considered highly sensitive indicators of climate change due to their narrow tolerance ranges for environmental conditions.

The Great Barrier Reef (GBR), located in the Coral Sea along the northeast coast of Australia, is the largest coral reef system in the world. Extending over 2,300 kilometers and consisting of more than 2,900 individual reefs, the GBR is recognized as a UNESCO World Heritage Site for its outstanding universal value. Beyond its ecological importance, the reef plays a critical economic role by supporting fisheries, coastal protection, and, most notably, tourism. Reef-related tourism contributes billions of dollars annually to the Australian economy and is a major source of employment in Queensland.

Despite its significance, the Great Barrier Reef has experienced substantial degradation over recent decades. Climate change has intensified environmental pressures on the reef, threatening both its ecological integrity and the economic activities that depend on it.

Understanding the interaction between climate-driven reef stress and tourism activity is therefore essential for sustainable environmental and economic planning.



Figure 1: Geographical Location of the Great Barrier Reef

Location of the Great Barrier Reef

Figure 1 presents the geographical location of the Great Barrier Reef along the northeast coast of Australia. The reef extends parallel to the coastline of Queensland in the Coral Sea and spans more than 2,300 kilometers. Its proximity to major coastal cities and accessibility via reef tourism hubs has made it one of the most visited marine ecosystems in the world.

Coral Reefs and Coral Bleaching:

Coral ecosystems:

Corals are living marine organisms composed of small animals called polyps, which form colonies and secrete calcium carbonate skeletons. Over long periods, these skeletons accumulate to create complex reef structures that support high levels of marine biodiversity. Corals maintain a symbiotic relationship with microscopic algae known as zooxanthellae, which reside within coral tissues. These algae perform photosynthesis, supplying corals with essential nutrients and giving reefs their characteristic coloration.

Coral bleaching:

Coral bleaching occurs when corals are exposed to environmental stress, most commonly elevated sea surface temperatures. Under prolonged thermal stress, corals expel their symbiotic zooxanthellae, causing the coral tissue to become transparent and revealing the white calcium carbonate skeleton beneath. While corals can survive short-term bleaching events, repeated or severe episodes significantly increase the likelihood of coral mortality.

Bleaching events reduce live coral cover, biodiversity, and the visual quality of reef environments. As a result, coral bleaching not only disrupts marine ecosystems but also diminishes the recreational and aesthetic value of reef destinations, potentially influencing tourist behavior and visitation patterns.

Key Terminology (Data Metrics) and Climate Stress Indicators:

To assess the impact of climate change on the Great Barrier Reef and its associated tourism activity, this project incorporates several climate and environmental indicators commonly used in coral reef monitoring:

- **Turbidity:** Diffuse Attenuation Coefficient at 490 nm. "The diffuse attenuation coefficient at 490 nm (K_d490) indicates the turbidity of the water column - how visible light in the blue to green region of the spectrum penetrates within the water column. The value of K_d490 represents the rate at which light at 490 nm is attenuated with depth." (https://eastcoast.coastwatch.noaa.gov/cw_k490.php)
- **Cyclone Frequency:** number of cyclone events from 1964 to 2014.
- **SST:** Climatological Sea-Surface Temperature based on weekly SSTs.
- **SSTA:** Sea-Surface Temperature Anomaly. It is the weekly SST minus weekly long-term mean SST.
- **SSTA Frequency:** number of times over the previous 52 weeks that $SSTA \geq 1$ degree Celsius.
- **SSTA DHW:** Sea Surface Temperature Degree Heating Weeks. It is the sum of previous 12 weeks when $SSTA \geq 1$ degree Celsius.
- **TSA:** Thermal Stress Anomaly. It is weekly SSTs minus the maximum of weekly long-term SSTs in degrees Celsius.

- **TSA Frequency:** The number of times over previous 52 weeks that $TSA \geq 1$ degree Celsius.
- **TSA DHW:** Thermal Stress Anomaly Degree Heating Weeks. It is sum of previous 12 weeks when $TSA \geq 1$ degree Celsius.
- **GSP:** Gross State Product (Queensland).
- **GDP:** Gross domestic product.

Sully et al. (2019) and Sully et al. (2020) showed that temperature metrics such as SST, TSA, thermal-stress indicators and turbidity make strong positive contributions to bleaching likelihood. These findings support the idea, emphasized by van Woesik et al. (2022), that temperature metrics can be used as reliable indicators for evaluating coral bleaching on coral reefs.

Relationship Between Reef Health and Tourism:

Tourism activity in the Great Barrier Reef region is closely linked to reef conditions. Healthy reefs characterized by high coral cover, biodiversity, and clear waters are major attractions for activities such as snorkeling, scuba diving, and reef cruises. Conversely, reef degradation caused by bleaching events, storm damage, and reduced water quality can negatively affect visitor satisfaction and perceptions, potentially leading to reduced visitation.

In addition to environmental conditions, tourism demand is influenced by broader economic factors. To account for these effects, this study includes Gross Domestic Product (GDP) and Gross State Product (GSP) as economic indicators representing national and regional economic activity.

Problem Statement and Research Objectives:

Although the ecological impacts of climate change on coral reefs are well documented, quantifying the associated economic consequences particularly those related to tourism—remains a complex and evolving challenge. The Great Barrier Reef provides a valuable case study for examining how climate-induced environmental stress translates into changes in tourism demand over time.

The primary objective of this project is to analyze and model the relationship between climate change indicators and tourism activity in the Great Barrier Reef region. Specifically, the project aims to:

1. Identify key climate and environmental variables associated with coral stress and reef degradation.
2. Evaluate the extent to which these variables influence tourism visitation, measured through overnight tourist trips.
3. Develop and compare predictive regression models to forecast tourism activity under varying climate conditions.

By integrating climate, environmental, and economic data with predictive analytics techniques, this study seeks to provide evidence-based insights into the long-term implications of climate

change for reef-based tourism. The findings aim to support informed decision-making for policymakers, environmental managers, and stakeholders involved in the sustainable management of the Great Barrier Reef.

Major Coral Bleaching Events in the Great Barrier Reef:

Over the past decade, the Great Barrier Reef has experienced several large-scale coral bleaching events linked to prolonged periods of elevated sea surface temperatures.

- **2016** - an extreme marine heatwave caused the most severe bleaching event ever recorded on the reef, resulting in widespread coral mortality, particularly in the northern sections.
- **2017** - marking one of the first instances of consecutive-year bleaching, which significantly reduced the reef's ability to recover.
- **2020** - A further widespread bleaching event occurred, reinforcing concerns that mass bleaching is becoming more frequent under ongoing climate warming.

These repeated bleaching events have reduced coral cover and altered reef composition, raising concerns about the long-term resilience of the reef and its capacity to sustain tourism-dependent activities.

Rationale for Predictive Modeling and Machine Learning:

The relationship between climate change, reef health, and tourism demand is complex and influenced by multiple interacting environmental and economic variables. Predictive modeling and machine learning techniques are well suited to this context as they allow for the integration of diverse datasets, identification of key drivers, and estimation of future tourism outcomes under changing climate conditions. Such approaches provide a data-driven framework for assessing potential impacts and supporting forward-looking decision-making.

Data

The data used for this project was obtained from two primary sources. Coral bleaching indicators, particularly thermal-stress metrics, were obtained from the database compiled by van Woesik et al. (2022). This dataset integrates 34,846 bleaching records from 14,405 sites in 93 countries, in a time frame between 1980 and 2020. It was generated from seven independent sources and standardized using information from Marine Ecoregions of the World (MEOW), Coral Reef Temperature Anomaly Database (CoRTAD), National Aeronautics and Space Administration (NASA) and Earth Observing System Data and Information System (EOSDIS). The final database consists of 20 related tables containing spatial, temporal, and environmental attributes.

Tourism data was sourced from Hyndman et al. (2021), as these records of overnight trips allowed the best temporal overlap with the thermal-stress database. This data represents the number of nights Australians spent away within the Great Barrier Reef (GBR) region.

Linking the two datasets was rather challenging. Even though the thermal-stress database offers good spatial resolution, integrating it with the overnight-trips database required aggregation, resulting in a combined dataset with quarterly temporal granularity and averaged thermal-stress metrics across the GBR.

Data Visualization

We can plot some of the variables in the data to see how they behave through time.

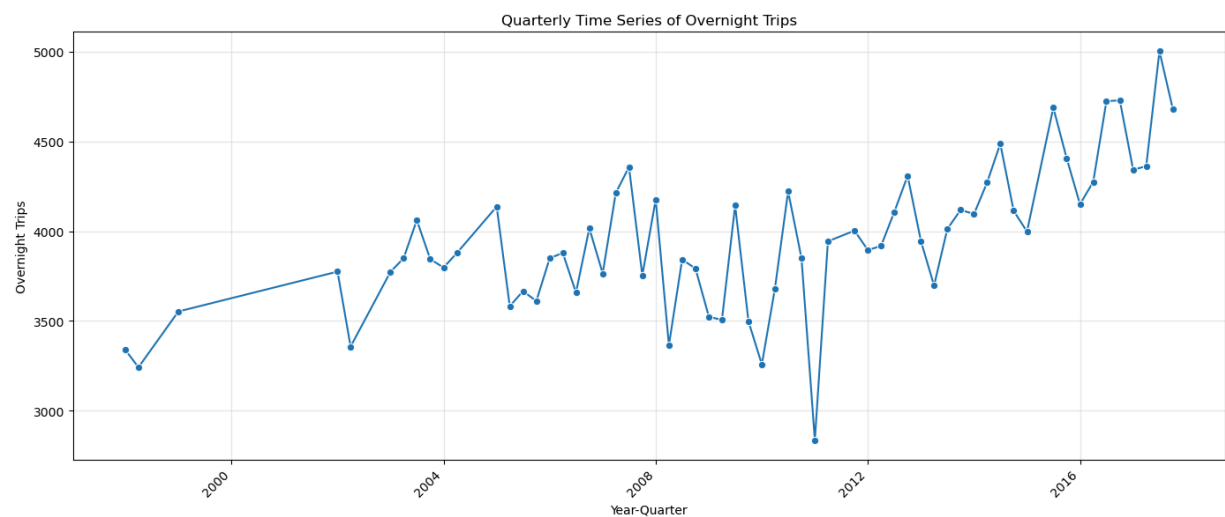


Figure 2: Target variable

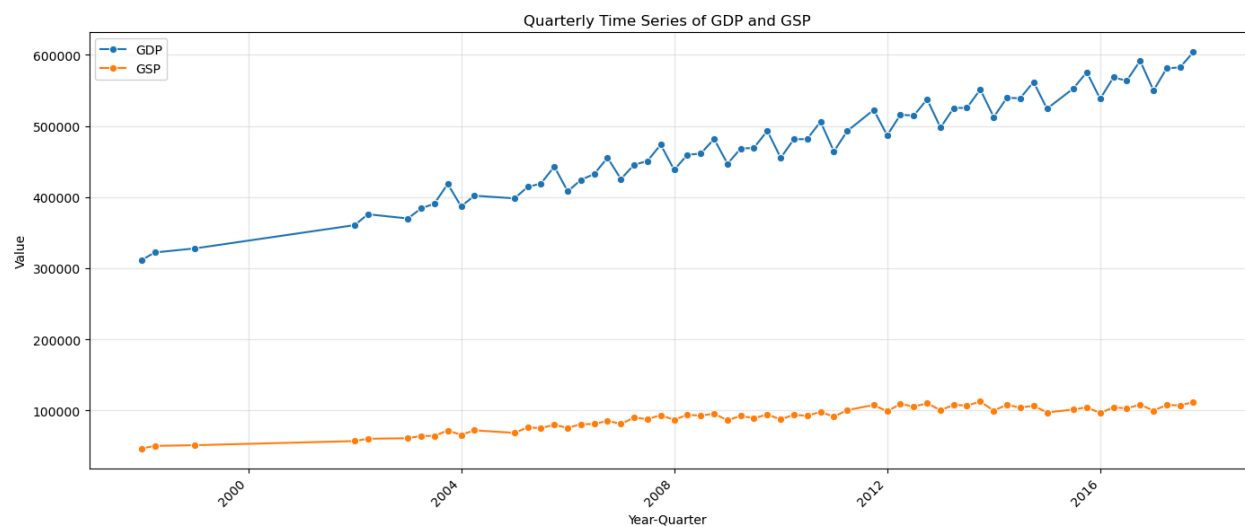


Figure 3: GDP and GSP

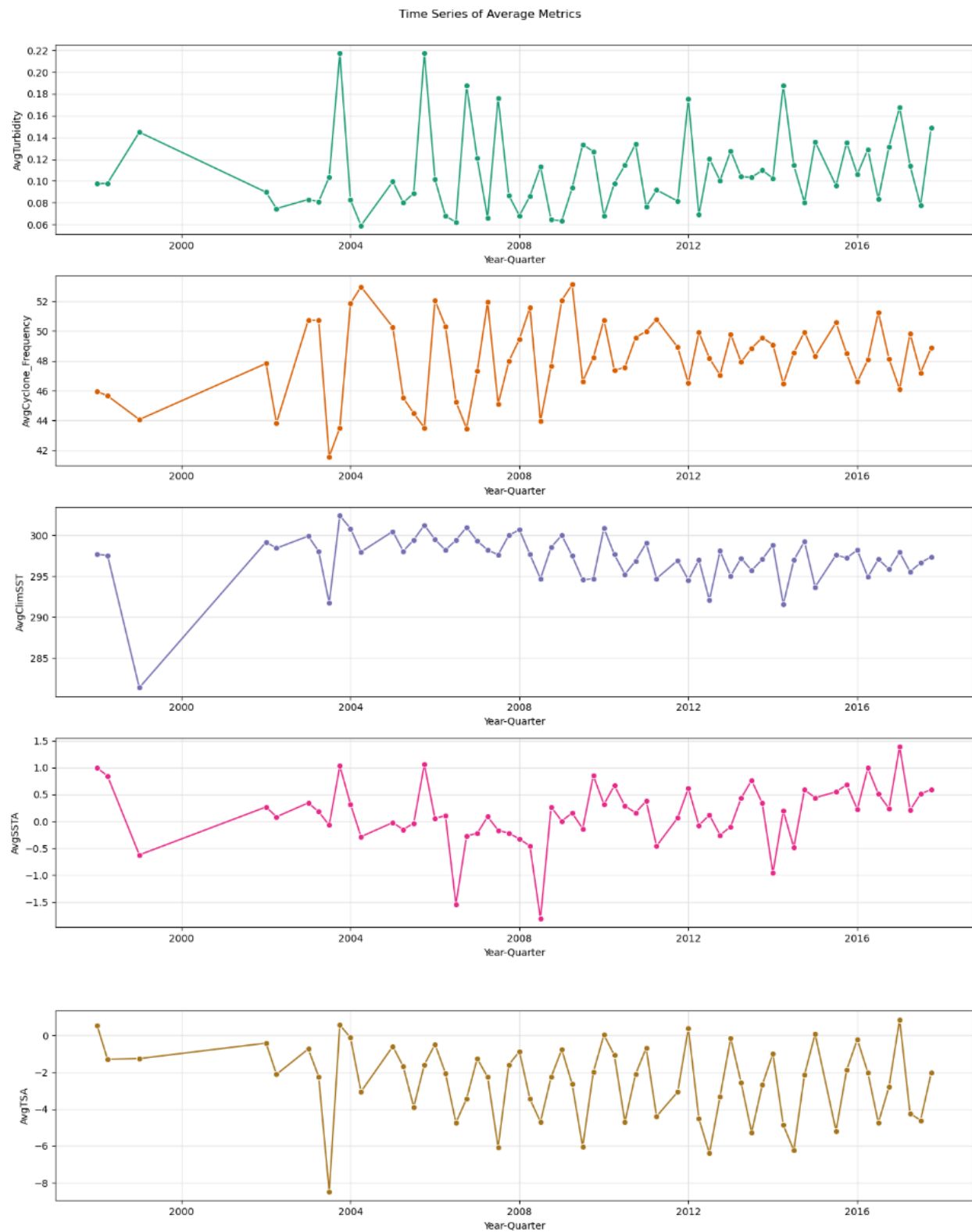


Figure 4: Thermal-stress indicators

Tourism prediction

The resulting combined dataset ranges from 1998 to 2017. We will use the data metrics previously described and overnight trips (overnight_trips) as a measure of tourism visitation. In other words, the thermal-stress metrics (along with GDP, GSP and turbidity), will be the features used to predict the overnight trips, as the target variable.

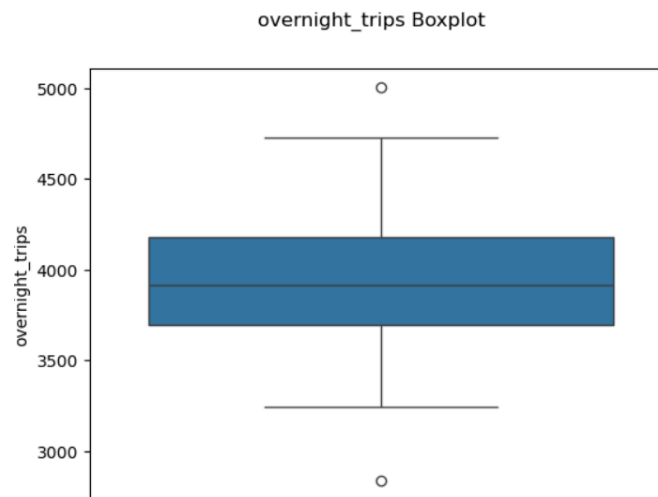


Figure 5: Target variable boxplot

In the boxplot we can see the distribution of the target variable. Considering its numerical nature, and that it seems well balanced without significant outliers, a regression task seems appropriate.

The following features were defined for modeling task, representing the time variable and the data metrics previously discussed: Quarters, AvgTurbidity, AvgCyclone_Frequency, AvgClimSST, AvgSSTA, AvgSTA_Frequency, AvgSSTA_DHW, AvgTSA, AvgTSA_Frequency, AvgTSA_DHW, GSP and GDP.

The number of features (12) is smaller than the number of observations (61); however, it is still a good idea to reduce complexity, increase interpretability and decrease the variance of the model by reducing the number of features. Therefore, we focus on the following models that incorporate implicit feature selection or regularization:

- Linear regression with Forward Stepwise feature selection
- Lasso of degree 2
- Lasso of degree 1
- Ridge of degree 1

Preprocessing

Before modeling, several preprocessing steps were defined. Categorical variables, specifically the “Quarters” feature, were encoded using one hot encoding, while numerical variables were standardized using standard scaling to ensure comparable scales.

Model evaluation was conducted using nested k fold cross validation with five folds and 80% of the data set to train set, allowing the model to be validated on multiple subsets and reducing correlations among validation outputs compared to other approaches. For Lasso and Ridge regression, the regularization parameter (alpha) was tuned using Grid Search procedure.

As a baseline, we start with a simple average model to provide a reference to compare against more complex models. The metric used for comparison is Mean Absolute Error.

```
mae_avg = mean_absolute_error(y_test, [y_train.mean()]*len(y_test))
print(mae_avg)
```

The result is a **MAE of 405.84**

Linear regression with Forward Stepwise feature selection

We begin with the Forward Stepwise feature selection process, using Mean Absolute Error as the criteria to choose a better feature subset.

```
all_features = list(features)
current_features = list() #start with empty model
best = Solution(y) #initialize solution

while current_features != all_features:
    selected_feature = None

    for feature in set(all_features) - set(current_features):
        new_features = current_features + [feature] #add one feature at a time
        maes = cross_val_score(
            estimator=get_model(new_features),
            X=X_train[new_features], y=y_train,
            cv=kfold, scoring='neg_mean_absolute_error')
        mae = -np.average(maes)

        if best.update(new_features, mae):
            selected_feature = feature

    if selected_feature:
        current_features.append(selected_feature)
    else:
        break

best_features_lin = best.features
print('Selected features: ', end='')
print(', '.join(best_features_lin))
print(f'MAE feature selection: {round(mae,4)}')
```

The resulting selected features are GDP, GSP and AvgTSA with a MAE of 236.08

The next step is to fit the linear regression model with this best feature subset:

```
categorical_features_lin = []
numerical_features_lin = []

for feature in best_features_lin:
    for cat_feature in categorical:
        if feature == cat_feature:
            categorical_features_lin.append(feature)
    for num_feature in continuous:
        if feature == num_feature:
            numerical_features_lin.append(feature)

linear_reg = make_pipeline(
    ColumnTransformer(transformers=[
        ('categorical', OneHotEncoder(), categorical_features_lin),
        ('numerical', StandardScaler(), numerical_features_lin)
    ]),
    LinearRegression())

# Train model with selected features
linear_reg.fit(X_train[best_features_lin], y_train)

# Make predictions
y_pred_lin = linear_reg.predict(X_test)

# Calculate MAE
print(f"MAE linear regression: {mean_absolute_error(y_test, y_pred_lin)}")
```

The result is a **MAE of 180.97**, with the following coefficients:

```
Coeff GDP: 571.0081443638201
Coeff GSP: -366.42551552641976
Coeff AvgTSA: -111.89472452983362
```

Examining the estimated coefficients shows that GSP and average TSA have a negative influence on the target variable. This means that increases in either metric are associated with a reduction in the predicted number of overnight trips.

Lasso degree 2

The next model applies Lasso regression with polynomial features of degree 2, which creates additional quadratic and interaction features to capture nonlinear relationships in the data. Since Lasso is a regularization method that shrinks coefficients toward zero, it naturally performs variable selection by keeping only the most influential predictors and setting others exactly to zero.

```

Lasso_reg_d2= make_pipeline(
    ColumnTransformer(transformers=[
        ('categorical', OneHotEncoder(), categorical),
        ('numerical', StandardScaler(), continuous)
    ]),
    PolynomialFeatures(degree=2),
    GridSearchCV(estimator=Lasso(max_iter=100000),
        param_grid=dict(alpha=np.logspace(-3, 5, 50)),
        cv=kfold, scoring='neg_mean_absolute_error')
)

# Train model with selected features
Lasso_reg_d2.fit(X_train, y_train)

# Make predictions
y_pred_lasso_d2 = Lasso_reg_d2.predict(X_test)

# Calculate MAE
mean_absolute_error(y_test, y_pred_lasso_d2)

```

The result is a **MAE of 267.06**, with an alpha of 54.28 and the following coefficients:

	feature	coefficient
11	numerical_AvgTSA	-15.675648
15	numerical_GDP	175.900533
89	numerical_AvgCyclone_Frequency numerical_GSP	-1.461966
100	numerical_AvgSSTA^2	-1.870917
111	numerical_AvgSTA_Frequency numerical_AvgTSA_...	55.859433
114	numerical_AvgSTA_Frequency numerical_GDP	43.171001
121	numerical_AvgTSA^2	19.786716
127	numerical_AvgTSA_Frequency numerical_AvgTSA_DHW	-23.092453
130	numerical_AvgTSA_DHW^2	-2.155508
135	numerical_GDP^2	16.127588

In this model we can observe a variety of interactions among features. Notably, GDP shows the strongest positive influence on the target prediction. In contrast, the interaction between TSA Frequency and TSA DHW shows the strongest negative effect, followed by the average TSA feature, both of which are associated with reductions in overnight trips prediction.

Lasso degree 1

The following model uses Lasso with polynomial features degree of 1, so it does not generate quadratic or interaction features.

```

Lasso_reg_d1= make_pipeline(
    ColumnTransformer(transformers=[
        ('categorical', OneHotEncoder(), categorical),
        ('numerical', StandardScaler(), continuous)
    ]),
    GridSearchCV(estimator=Lasso(max_iter=100000),
        param_grid=dict(alpha=np.logspace(-3, 5, 50)),
        cv=kfold, scoring='neg_mean_absolute_error')
)

# Train model with selected features
Lasso_reg_d1.fit(X_train, y_train)

# Make predictions
y_pred_lasso_d1 = Lasso_reg_d1.predict(X_test)

# Calculate MAE
mean_absolute_error(y_test, y_pred_lasso_d1)

```

The result is a **MAE of 260.24**, with an alpha of 54.28 and the following coefficients:

	feature	coefficient
10	numerical_AvgTSA	-45.058258
14	numerical_GDP	180.095221

This model shows coefficient signs consistent with previous results: GDP continues to make a positive contribution, while the average TSA remains negative. This shows a pattern suggesting that higher economic activity is associated with increased overnight trips prediction, while rising thermal-stress anomalies correspond to a decline in overnight trips prediction, which could be linked to the elevated risk of coral bleaching under higher TSA conditions.

Ridge degree 1

In Ridge regression coefficients do not really reach zero, therefore it is not considered a variable selection method, but we can assess if a feature is less relevant compared to the others.

```

Ridge_reg= make_pipeline(
    ColumnTransformer(transformers=[
        ('categorical', OneHotEncoder(), categorical),
        ('numerical', StandardScaler(), continuous)
    ]),
    GridSearchCV(estimator=Ridge(max_iter=100000),
        param_grid=dict(alpha=np.logspace(-3, 5, 50)),
        cv=kfold, scoring='neg_mean_absolute_error')
)

# Train model with selected features
Ridge_reg.fit(X_train, y_train)

# Make predictions
y_pred_ridge = Ridge_reg.predict(X_test)

# Calculate MAE
mean_absolute_error(y_test, y_pred_ridge)

```

The result is a **MAE of 300.19** , with an alpha of 79.06 and the following coefficients:

	feature	coefficient
0	categorical_Quarters_Q1	-1.177796
1	categorical_Quarters_Q2	-17.830185
2	categorical_Quarters_Q3	20.465377
3	categorical_Quarters_Q4	-1.457395
4	numerical_AvgTurbidity	13.048370
5	numerical_AvgCyclone_Frequency	1.960834
6	numerical_AvgClimSST	-15.996971
7	numerical_AvgSSTA	6.610007
8	numerical_AvgSTA_Frequency	15.606841
9	numerical_AvgSSTA_DHW	-2.607519
10	numerical_AvgTSA	-34.880844
11	numerical_AvgTSA_Frequency	3.911152
12	numerical_AvgTSA_DHW	-14.674613
13	numerical_GSP	38.775185
14	numerical_GDP	67.570966

We can observe several coefficients in this model that are close to zero, such as Cyclone Frequency, SSTA DHW, and the categorical features for quarter 1 and quarter 4, suggesting that these variables have less influence on the target. However, even with these weaker features included, the resulting MAE is relatively high compared to previous models, indicating that the added features introduce some additional variance without improving predictive performance.

The Mean Absolute Error of the different models can be summarized in the following table:

Model	Mean Absolute Error
Average	405.84
Linear regression with FS selection	180.97
Lasso degree 2	267.06
Lasso degree 1	260.24
Ridge degree 1	300.19

When evaluating the performance based on the Mean Absolute Error, the best model was Linear regression with Forward Stepwise feature selection.

Possible effects on tourism

We can quantify possible losses in tourism by creating an artificial dataset with expected metrics. The 4 quarters of 2018 are set in a new dataset with expected metrics provided by NOAA (Skirving, W. J. et al. 2020), and overnight trips forecasted by the Australian Trade and Investment Commission (2021).

We selected the linear regression model to generate predictions for the new dataset, as it achieved the lowest MAE among all the models evaluated. In the following table we can see the model prediction results, the forecasted trips by the Australian Trade and Investment Commission (y_{exp}), and the difference in percentage compared to the actual overnight trips in 2017 (% Diff with 2017):

	y_{exp}	model prediction
0	4721.458084	4265.328577
1	4743.702688	4599.470039
2	5439.643404	4735.525568
3	5090.796824	4646.766969
Total	19995.601000	18247.091154
% Diff with 2017	8.740000	-0.770000

As we can see, the Australian Trade and Investment Commission projected an increase in overnight trips relative to 2017 of 8.74%. However, our prediction model estimates a decline of 0.77%. In a similar analysis in the Indian Ocean, Westmacott et al., (2000) have assumed a link between this pattern of decline in trips and coral bleaching, highlighting its potential influence on visitor behavior.

Limitations of the Current Study

1. Limited temporal coverage and sample size

The dataset used for model fitting spans from 1998 to 2017, resulting in 61 quarterly observations. While this period captures important bleaching-related climate variability, the relatively small number of observations limits the statistical power of the models and constrains the complexity of methods that can be reliably applied. In addition, direct measurements of coral bleaching severity or coral cover loss are unavailable, requiring reliance on proxy indicators instead of direct response variables. Under such conditions, more flexible or high-capacity machine learning models are prone to overfitting, reducing their reliability and generalizability.

2. Tourism indicator is indirect

Tourism activity is represented by overnight trips, which is a broad indicator of tourism demand in the Great Barrier Reef region. However, this metric does not exclusively capture reef-dependent tourism activities such as snorkeling, scuba diving, or reef cruises. As a result, changes in overnight trips may also reflect non-reef-related travel motivations, potentially diluting the direct effect of coral reef degradation on tourism behavior.

4. Economic confounding effects

Tourism demand is strongly influenced by macroeconomic conditions, as reflected by the importance of GDP and GSP in the selected models. Although these variables help control economic trends, they may also dominate the contribution of climate-related variables, making it difficult to isolate the marginal effect of reef stress on tourism demand.

Future Work

1. Explicit modeling of lagged and cumulative effects

Introduction of lagged climate and bleaching variables (e.g., previous-quarter or previous-year stress indicators) and cumulative exposure measures. This would better reflect the delayed and persistent impacts of reef degradation on tourist perceptions and travel decisions.

2. Advanced time-series validation strategies

Applying time-series cross-validation or rolling-window forecasting techniques would provide a more realistic evaluation of predictive performance compared to random data splits. This would strengthen the credibility of future tourism forecasts under changing climate conditions.

3. Causal impact analysis around bleaching events

Beyond predictive modeling, future research could explore causal inference approaches, such

as interrupted time series analysis around major bleaching events (e.g., 2016, 2017, 2020), to better isolate the causal impact of bleaching on tourism demand.

Conclusion

This study explored the relationship between climate-related environmental stress indicators and tourism activity in the Great Barrier Reef region using quarterly data from 1998 to 2017. By integrating climate stress proxies, economic indicators, and tourism demand measures, the analysis provides evidence of associations between environmental conditions and tourism outcomes.

Importantly, the findings should be interpreted as **exploratory rather than causal**. Tourism demand in the region is influenced by a complex interaction of economic, environmental, and social factors, and the available data do not allow for full causal identification of bleaching impacts. Nevertheless, the results suggest that climate stress indicators contribute to explaining tourism variability beyond purely economic drivers.

The observed dominance of macroeconomic variables such as GDP and GSP highlights the challenge of isolating environmental effects in aggregate tourism data. At the same time, the inclusion of thermal stress variables improves model performance, indicating that climate-related reef stress carries measurable informational value.

Overall, this study demonstrates the potential of predictive analytics as a tool for exploring climate–tourism relationships, while also underscoring the need for richer reef health data, longer time series, and causal modeling approaches in future research.

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